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COLLEGE OF COMPUTER STUDIES

BRGYALERT: A RESPONSIVE INCIDENT REPORTING AND ANALYTICS PLATFORM FOR BARANGAY

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CHAPTER I
INTRODUCTION

Background of the Study

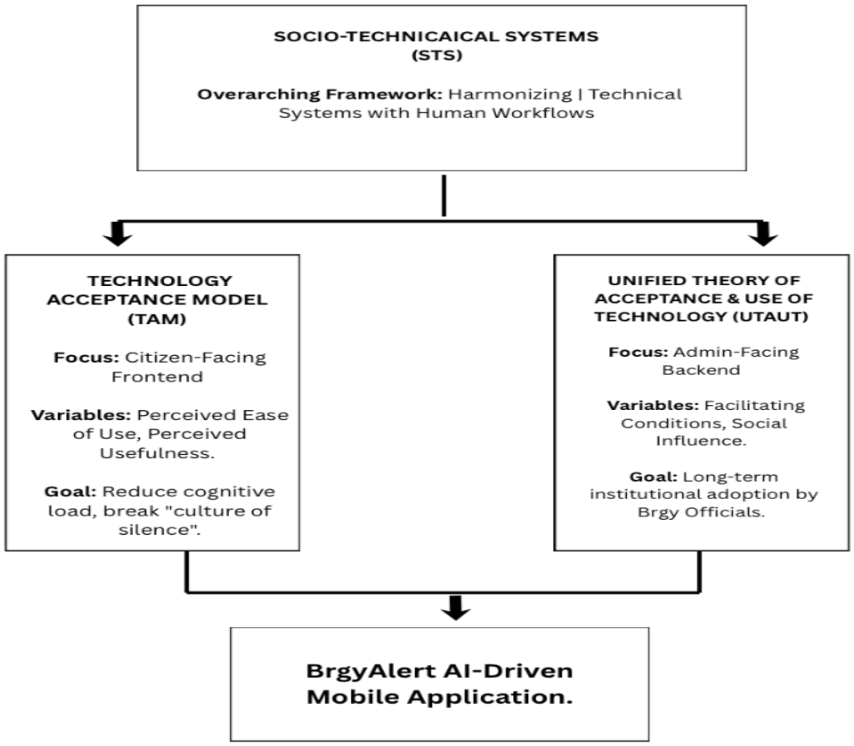
A digital transformation in the current era of governance is no longer an option; it is a necessity. In the Philippines the barangay is the main administrative arm that takes care of the pressing needs of the community and settles disputes. Department of the Interior and Local Government (DILG) Memorandum Circular No. 2017-144 says every barangay must have a Violence Against Women (VAW) desk to handle sensitive domestic issues. Nevertheless, many barangays are still using old-fashioned paper-based filing systems that are vulnerable to physical damage, unauthorized access, and long delays in data retrieval, despite these regulations.

The situation created from manual record keeping is described by scholars as a “socio-technical void” where the resources of local officials are inadequate to cope with the increasing complexity of contemporary event reporting. There are centralized systems, for example, PNP-CIRAS for national law enforcement, but there is a conspicuous lack of localized, user-friendly interfaces made specifically for the grassroots level. This gap often results in poor recordkeeping and an inability to track recurring events that could develop into more dangerous risks. Furthermore, the advent of automated risk assessment and algorithmic policing has shown that data can do more than just store information; it can also predict and prevent harm. In the context of domestic violence and community safety, systems that can automatically identify high-risk instances are necessary for trauma-informed governance. Without an intelligent system that can classify these reports, barangay officials are left to manually prioritize cases based on their subjective assessment, potentially missing warning indicators of increasing violence.

The authorities in Barangay Lepa are usually on the move, responding to calls, on patrol, and helping residents outside the office. These responders are limited to a workstation when using a web-based or paper-bound system and cannot log and access information in real time. This lack of mobility often leads to "delayed reporting," where facts of the incident are lost or forgotten before it is formally reported at the barangay hall.

In order to address these issues, the researchers plan to create BrgyAlert, a mobile application tailored for Barangay Lepa. With a focus on a mobile-only platform, the system provides portable and accessible incident management at the point of need. This AI-powered risk stratification system enables real-time analysis of the reported data and identification of high-risk situations, giving officials a data-driven basis for prioritization.

The study aims to revive barangay governance through this mobile ecosystem by connecting the gap between administrative duty and technological efficiency.



Theoretical Framework

The development of the BrgyAlert system is grounded in several established theoretical frameworks that bridge the gap between technological innovation and social governance.

Figure 1.1 Brgyalert mobile incident management system

Theoretical Synthesis and Conceptualization

The BrgyAlert system's development is based on a number of well-known theoretical frameworks that connect social governance with technical innovation. The Socio-Technical Systems (STS) Theory, which holds that any organizational unit's efficacy is determined by the dynamic interplay between its social components and its technical tools, serves as the main basis for this study. The people and local officials make up the social system in the setting of Barangay Lepa, whilst the mobile application and its integrated artificial intelligence represent the technical system. By using STS Theory, this study makes sure that the AI-driven risk classification works as a decision-support tool that supports the barangay's legislative mandates, including the establishment of the Violence Against Women (VAW) Desk, rather than operating in a vacuum. In order to keep the system attentive to the community's trauma-informed requirements, technology and human intervention must work in harmony.

The Technology Acceptance Model (TAM), which emphasizes how each individual user interacts with the mobile platform, complements STS's organizational viewpoint. According to TAM, the adoption of BrgyAlert is determined by two key factors: perceived usefulness (PU) and perceived ease of use (PEOU). The system's "Perceived Usefulness" is based on its capacity to use AI to automate the process of prioritizing high-risk cases, which was previously done manually and subjectively. The study's mobile-only methodology addresses "Perceived Ease of Use" in the meanwhile. The solution reduces the barrier to entry and makes digital reporting more accessible than conventional, desk-bound ways by leveraging a platform that is already essential to the everyday lives of both the administrators and the constituents.

Furthermore, the study employs the Unified Theory of Acceptance and Use of Technology (UTAUT) to address actual facilitating conditions in the local context. According to UTAUT, a technology rollout's effectiveness is contingent upon the organization's social influence and resource availability. This study's exclusive focus on a mobile application is in line with Barangay Lepa's current infrastructure, where smartphone usage is strong and authorities are regularly on the go. The shift from a manual, paper-based log to a portable, intelligent ecosystem is justified by this theoretical approach, guaranteeing that the technology is both practical and helpful for the actual limitations of grassroots governance.

Conceptual Framework

The Conceptual Framework of this study is structured using the Input-Process-Output (IPO) model, which provides a systematic roadmap for transforming raw requirements into a functional mobile solution.

Figure 1.2 Conceptual Framework of the BrgyAlert Mobile Application.

The Input stage includes the fundamental elements required to construct BrgyAlert, which are divided into knowledge, technological, and data requirements. This involves a thorough examination of Barangay Lepa's current incident reporting procedures as well as the legislative requirements of Republic Act No. 9262, often known as the Anti-Violence Against Women and Their Children Act. Technically, the inputs include the hardware specs of smartphones used by locals and barangay authorities, as well as the software development kits (SDKs) required for the creation of mobile applications. These inputs guarantee that the system is based on a thorough understanding of the community's technical environment as well as the legal framework.

The Process stage describes the iterative development lifecycle used to bridge the gap between the initial requirements and the finished product. To make sure the application meets the unique mobility demands of the barangay staff; this phase starts with methodical requirements collecting and stakeholder consultations. The creation of a mobile-optimized User Interface (UI) and User Experience (UX) that prioritizes usability for constituents under pressure comes next. The technical development of the AI risk stratification algorithm, which entails training models to classify incident reports according to severity, is the focus of this phase. During this stage, thorough system testing and assessment are also carried out to confirm the precision of the AI's flagging mechanism and the general stability of the mobile platform.

This output is a portable ecosystem intended for real-time community reaction, in contrast to conventional paper-based or web-tethered systems. The application's main feature is its capacity to automatically rank high-risk home and communal occurrences, giving officials quick, data-driven insights. The technology enables quicker response times and more precise data management for the Barangay Lepa Violence Against Women (VAW) Desk by combining these reports into a secure mobile interface. The system's evaluated performance results, which show its efficacy in a real-world local governance situation, are also included in the final product.

Statement of the Problem

Barangay Lepa’s Violence Against Women (VAW) Desk relies on manual blotter records, creating a strictly reactive system with fragmented data, slow Barangay Protection Order

(BPO) processing, and high risks of record tampering. Without discreet, safe reporting options, victims fear immediate retaliation and social stigma, reinforcing a dangerous "culture of silence" and discouraging them from reporting abuse.

To address these compounding security and administrative challenges, this study develops BrgyAlert: A Mobile-Based Incident Management and AI-Driven Risk Stratification System. The system integrates an accessible mobile interface with an Artificial Intelligence (AI) engine to optimize the prioritization of cases and ensure secure data management.

Specifically, this study aims to answer the following research questions:

1. What are the functional, non-functional, and security requirements of the BrgyAlert mobile application based on the current field challenges of the Barangay Lepa Violence Against Women (VAW)Desk?

   1.1. What are the specific user requirements for a discreet and responsive mobile reporting interface for victims?

   1.2. What administrative prerequisites must barangay officials meet in order to use an automated analytical dashboard?

1.3. What encryption and data security procedures are necessary to guarantee the confidentiality and integrity of incident reports?

2. How can local incident reporting and data processing be optimized in the design and development of the BrgyAlert system?

   2.1. How will the Artificial Intelligence (AI) engine be designed to classify situational risk urgency and analyze behavioral data inputs?

   2.2. How will the secure database of the application be set up to guarantee data integrity and stop illegal access?

3. What is the level of acceptability of the BrgyAlert application among its primary stakeholders in terms of TAM variables?

   3.1. What is the mobile interface's perceived ease of use (PEOU) for barangay employees and community reporters?

   3.2. What is the AI risk stratification engine's perceived usefulness (PU) in terms of expediting protective order workflows and improving community safety?

4. How does the developed BrgyAlert mobile application improve the efficiency of incident documentation and the processing time of Barangay Protection Orders (BPOs) compared to the traditional manual blotter system?

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Assumptions of the Study

Technical and Operational Assumptions. It is assumed that the main stakeholders in Barangay Lepa, such as community members and barangay authorities, have regular access to cellphones. It is assumed that mobile internet connectivity is available for AI-assisted functions, because BrgyAlert performs text analysis and risk stratification through a cloud-assisted AI service rather than as a fully offline native JavaScript NLP engine. It is also assumed that the secure cloud backend and Firebase services are accessible for real-time data synchronization, incident logging, and AI processing. And lastly, it assumed that offline incident submission is still possible through an SMS fallback channel, but that AI categorization, summary generation, and report validation require network access.

Behavioral and Stakeholder Assumptions. The study assumes that the local government unit and the Barangay Council of Barangay Lepa are willing to move from traditional paper-based blotter processes to a digital-first environment. It is assumed that barangay officials and VAW Desk personnel will engage with the system and consider its AI risk stratification output as a useful decision-support tool. It is believed that community members using the discreet mobile reporting interface will provide authentic, real-time incident descriptions and genuine evidential logs. It is anticipated that during Technology Acceptance Model (TAM) testing and system performance assessments, all parties participating in the evaluation phase will provide truthful, accurate, and objective feedback.

Evaluative and Analytical Assumptions. It is assumed that Barangay Lepa's historical data on the manual blotter system's processing timeframes is adequately documented to function as a trustworthy baseline for comparative efficiency analysis. It is assumed that the research team can collect sufficient evaluation data from barangay officials and community representatives to assess usability and effectiveness. It is considered that the system developers and IT specialists chosen to assess the platform have the technical know-how and subject-matter expertise required to effectively evaluate the system's performance and cloud-based AI architecture in accordance with industry standards

Scope and Delimitation

This study focuses fundamentally on the design, development, and implementation of BrgyAlert: A Mobile-Based Incident Management and AI-Driven Risk Stratification System, explicitly engineered for and deployed within the micro-local governance structure of Barangay Lepa.

The application's operational scope recognizes three distinct user classifications: community residents or victims who need a discreet mobile tool to report domestic and community incidents; Barangay officials and Violence Against Women (VAW)Desk personnel who manage case data, view analytics, and issue administrative dispatches; a system administrator responsible for maintaining the digital infrastructure.

In order to automate the triage process, the system uses artificial intelligence (AI) to classify situational risk categories (e.g., Low, Medium, or High) and parse event descriptions. The mobile application is implemented in React Native and integrates with a secure cloud backend. AI inference is performed through a cloud-assisted service using Google Gemini, not as a fully offline native JavaScript NLP engine on the device. Offline reporting is supported through an SMS fallback channel, but AI processing requires network connectivity. Strictly adhering to Republic Act No. 9262 requirements, this automation aims to maximize the pace at which Barangay Protection Orders (BPOs) are documented and issued. A mobile-responsive interface designed for low-bandwidth conditions makes up the system architecture, guaranteeing accessibility for users with different smartphone specifications. For data retention and real-time synchronization between reporters and responders, the system uses Firebase as its secure backend.

This study is limited to Barangay Lepa's official boundaries; reports from outside this area will not be ingested or analyzed by the system. BrgyAlert does not dynamically sync in real time with the database of the Philippine National Police (PNP); rather, it is designed as a grassroots intervention tool that is isolated from higher-echelon national law enforcement systems. Additionally, the system is technically constrained to using pre-existing cellular networks rather than developing physical network nodes or specialized communication infrastructure.

Significance of the Study

The development and implementation of the BrgyAlert application offer significant advancements in localized public administration and civic welfare. By transitioning grassroots incident tracking from a vulnerable, paper-based format to an automated and secure mobile system, this study provides distinct benefits to the following stakeholders:

Community Residents and Victims of Abuse. This study is extremely important to residents of Barangay Lepa, particularly those who are enduring domestic crises under Republic Act No. 9262. The platform offers a secure communication channel that avoids social shame and the danger of immediate escalation of retaliation by offering a covert and private mobile interface. As a result, victims are empowered to overcome the "culture of silence," knowing that their reports are immediately recorded and given priority by system triggers.

Barangay Officials and Violence Against Women (VAW) Desk Personnel For local administrators. This study transforms daily workflows from a reactive to a proactive, data-driven approach. Violence Against Women (VAW) Desk operators may quickly identify high-severity instances thanks to the AI-driven risk stratification engine, which removes the administrative inefficiencies associated with human log reviews. The prompt issuance of life-saving Barangay Protection Orders (BPOs) and the dynamic prioritization of emergency dispatches depend heavily on this efficienc

The Barangay Council of Barangay Lepa. At the local level, this system provides the legislative body with an analytical overview of safety metrics and geographical crime patterns. Local legislators can make well-informed decisions about the precise distribution of barangay resources, the tactical deployment of tanods (community watchmen), and the creation of focused public safety initiatives because to this actionable intelligence.

Software Engineers and System Designers. This study offers a useful paradigm for incorporating Natural Language Processing (NLP) into a low-bandwidth municipal platform to the technological development community. It demonstrates that sophisticated analytical tools may be effectively implemented using common mobile technologies and provides a roadmap for deploying AI triage structures inside contexts with limited resources.

Future Researchers. Finally, this study serves as a valuable academic reference for future researchers exploring the intersections of civic technology and grassroots governance. The methodologies and the theoretical integration of the Technology Acceptance Model (TAM) used in this study can be extended to design larger-scale smart municipal architectures for other local government units in the Philippines.

Definition of Terms

To ensure a unified understanding of the technical, administrative, and legal concepts used throughout this research, the following terms are operationally and conceptually defined based on how they are utilized within the BrgyAlert ecosystem:

AI Risk Stratification Engine The application's computational component parses and analyzes text inputs from event reports using algorithmic models. Its goal is to help administrators prioritize emergency actions by automatically classifying a situation's level of urgency (e.g., Low, Medium, or High Risk).

Barangay Blotter Accidents, crimes, and civil complaints in the community are recorded in the official record book utilized by local authorities. The Barangay Lepa Violence Against Women (VAW) Desk's manual, paper-based logging system, which the BrgyAlert system seeks to digitize, is represented in this study.

Barangay Protection Order (BPO) In accordance with Republic Act No. 9262, the Punong Barangay (Barangay Captain) enacted a statutory emergency relief measure. It commands an offender to stop threatening or carry out violent crimes against a victim right away.

BrgyAlert Mobile Application The main software product of this research is the BrgyAlert mobile application. It is a mobile-based system that includes an analytics dashboard for barangay officials to manage and prioritize reported incidents, as well as a reporting interface for community members.

Culture of Silence The sociological phenomenon where victims of domestic abuse choose not to report their experiences to local authorities due to social stigma or fear of retaliation. The system seeks to dismantle this barrier through discreet and confidential reporting channels.

Natural Language Processing (NLP) A subfield of artificial intelligence that allows the system to comprehend and interpret human language. The unstructured content in incident reports is parsed in this study using natural language processing (NLP) in order to provide data into the risk classification engine.

Perceived Ease of Use (PEOU) This is defined as the extent to which community members and barangay authorities feel that using the BrgyAlert mobile interface involves little mental or physical effort. It is derived from the Technology Acceptance Model (TAM).

Perceived Usefulness (PU) This is described as the extent to which stakeholders think using the BrgyAlert application directly improves their personal safety or speeds up administrative procedures. It is also derived from the Technology Acceptance Model (TAM).

VAW Desk (Violence Against Women and Their Children Desk) The front-line, specialized service center in Barangay Lepa is required by Philippine legislation to handle incidents of domestic abuse. It is the main location where the BrgyAlert administration interface is implemented.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter provides a detailed analysis of local and foreign literature and studies that serve as the theoretical and empirical underpinning for the development of BrgyAlert. The challenges of localized incident reporting in the Philippine barangay context, the integration of artificial intelligence in public safety analytics, the protocols for secure data management in sensitive reporting systems, and the socio-technical factors influencing user adoption of municipal e-governance platforms are the main topics covered by the thematic sections of the literature. The chapter identifies the main technological gaps that this research aims to fill by synthesizing these previous achievements.

Related Literature

Foreign Literature

The global transition toward digital public governance is framed not only as a technical upgrade but as a complex behavioral shift. Adiyono, Nurfaizah, and Sholikhah (2024) utilize the Unified Theory of Acceptance and Use of Technology (UTAUT) to argue that the success of such systems depends heavily on "facilitating conditions"—specifically internet stability and institutional support. For a platform like BrgyAlert, this suggests that the technical architecture must be resilient enough to operate in environments with varying connectivity. Complementing this, Ssaho (2024) emphasizes that digital modeling for municipal use must prioritize structural simplicity. He posits that complex administrative software often results in "cognitive overload" for local officials, thereby advocating for minimalist designs that ensure usability among non-technical users.

In the realm of security and intelligence, Novitzky, Kok, and Seneviratne (2023) highlight the evolution of "algorithmic policing." Their work explores how AI can analyze subtextual behavioral markers within emergency reports to detect high-risk patterns that human dispatchers might overlook due to volume or fatigue. However, this advancement brings ethical risks. Levy and Snell (2020) introduce the concept of "digital coercive control," warning that technology can be weaponized by abusers to monitor victims. This necessitates that safety applications implement high-level encryption and "stealth" reporting features to maintain victim confidentiality. Finally, Zuboff (2023) discusses "predictive visibility," suggesting that proactive monitoring allows local governments to transform from passive record-keepers into active participants in community safety, using data to allocate resources before a crisis escalates.

Local Literature

In the Philippine context, the foundation of community-based safety is rooted in Republic Act No. 9262 (Anti-VAWC Act) and DILG Memorandum Circular No. 2017-144, which institutionalize the Barangay VAW Desk. Despite these legal mandates, Santiago and Dela Cruz (2022) observe a persistent "culture of silence" in rural regions. They argue that geographical isolation and the lack of discreet reporting channels prevent victims from pursuing a Barangay Protection Order (BPO).

This issue is exacerbated by what researchers term "analog lag." Most barangays still rely on manual, paper-based Barangay Blotters, which Weger and Yeazitzis (2023) characterize as a "socio-technical void." They argue that manual tools are no longer capable of managing the complexity and urgency of modern crime data, leading to fragmented records that are prone to physical damage. To bridge this gap, any digital intervention must comply with the Data Privacy Act of 2012 (RA 10173). This law serves as the legal "guardrail" for BrgyAlert, ensuring that the transition from paper to cloud storage does not compromise the sensitive personal information of residents.

Related Studies

Foreign Studies

Empirical research provides a technical blueprint for the features within BrgyAlert. In the United Kingdom, Anderson and Miller (2022) demonstrated that machine learning algorithms can classify domestic violence severity with high accuracy by analyzing the frequency and nature of historical incidents. This provides the empirical basis for AI-driven triage in incident management. Similarly, a study in Singapore by Chen et al. (2021) found that "Effort Expectancy"—the user's perception of how easy a tool is to use—is the strongest predictor of adoption for e-governance apps.

The integrity of reported data is addressed by Braun and Schmidt (2023) in Germany, whose study confirms that advanced hashing algorithms and secure cloud architectures are essential for preserving the "forensic trail." This ensures that digital records are not just for storage but are admissible as evidence in legal proceedings. In the United States, Lee and Kim (2023) found that AI risk assessments can significantly reduce human bias during high-volume emergency triage, ensuring that cases are prioritized based on objective danger rather than subjective judgment. Furthermore, Patel et al. (2022) conducted research in India on "silent" apps, discovering that disguised UI elements (such as masking a safety app as a calculator or news feed) significantly increased reporting rates among populations living with their abusers.

Local Studies

Local studies emphasize the need for "granular" solutions. Aguirre and Manzano (2023) critiqued centralized systems like the PNP-CIRAS, noting that while they are excellent for national statistical gathering, they lack the localized usability needed for barangay-level responders who handle grassroots conflicts. This is supported by Bautista (2023), whose research indicates that barangay officials have a strong preference for mobile-based solutions over desktop dashboards. Since officials are often on patrol or responding to calls in the field, mobile accessibility is a prerequisite for real-time data entry.

Trust remains a critical barrier to adoption in the Philippines. Jou, Dela Cruz, and Santos (2024) found that data confidentiality and user awareness are the primary drivers of digital trust in government-led applications. If users do not feel their data is safe, they will return to manual reporting. Furthermore, Reyes and Mercado (2022) evaluated existing VAW Desks and found that "response lag" was primarily caused by disorganized manual filing systems rather than a lack of personnel. This proves that an analytics dashboard could theoretically halve response times. Finally, Lopez (2024) confirmed that using Agile methodology in Philippine public sector projects leads to higher success rates, as it allows for iterative feedback from the actual barangay captains and residents who will use the system.

Synthesis of the Review

The literature and studies reviewed provide a comprehensive roadmap for the development of BrgyAlert, bridging the gap between global technical standards and the specific socio-technical needs of the Philippine barangay.

A primary theme across the foreign literature and studies is the shift from reactive to proactive public safety through Artificial Intelligence (AI) and predictive analytics. Research by Anderson and

Miller (2022) and Novitzky et al. (2023) demonstrates that AI can identify linguistic risk markers and historical patterns in domestic violence that human responders might miss. However, for such a system to be legally viable, Braun and Schmidt (2023) and Levy and Snell (2020) emphasize that secure cloud architectures and encryption are non-negotiable to maintain forensic integrity and protect victims from "digital coercive control."

Locally, the literature highlights a significant "analog lag" within the Philippine barangay system. While RA 9262 and DILG mandates provide a legal framework for VAW Desks, Santiago and Dela Cruz (2022) and Weger and Yeazitzis (2023) identify a "socio-technical void" where manual paper-based blotters lead to data fragmentation and a "culture of silence" in rural areas. This justifies the transition toward a digital ecosystem that can handle the complexity of modern crime.

Regarding adoption, the Unified Theory of Acceptance and Use of Technology (UTAUT) serves as the theoretical anchor. Both foreign studies in Singapore (Chen et al., 2021) and local studies by Jou et al. (2024) confirm that for an e-governance tool to succeed, it must prioritize "Effort Expectancy" (ease of use) and "Data Confidentiality." Bautista (2023) further narrows this down to the Philippine context, noting that barangay officials specifically require mobile-based interfaces for real-time reporting during patrols.

Ultimately, BrgyAlert synthesizes these findings by integrating high-level AI risk stratification with a localized, mobile-responsive design. By addressing the "granular usability" missing in centralized national systems (Aguirre & Manzano, 2023) and utilizing Agile development for iterative community feedback (Lopez, 2024), this study aims to provide a secure, efficient, and culturally grounded solution for incident management in Barangay Lepa.

CHAPTER III

METHODOLOGY

This chapter highlights the systematic approaches and processes for the development and evaluation of BrgyAlert. It discusses the AI and mobile platform's technological requirements, the study design, the system development framework, and the statistical methods to test the system's effectiveness in speeding up incident reporting and risk management in Barangay Lepa.

The main goal of this methodology is to facilitate a structured transition from the traditional manual blotter system to a digitalized, AI-driven mobile ecosystem. The next sections describe the research environment, the technical requirements for AI risk stratification, and the metrics used to assess the application's functionality and user acceptance

Research Design

This study utilizes a Descriptive-Evaluative Research Design to guide the development and assessment of BrgyAlert: A Mobile-Based Incident Management and AI-Driven Risk Stratification System. The descriptive component is employed to identify and analyze current challenges in incident reporting within Barangay Lepa, such as the inefficiencies of manual blotter systems, risks associated with physical record storage, and delays in emergency response. Qualitative insights gathered from interviews with barangay officials serve as the foundation for defining the specific functional and technical requirements of the platform.

The evaluative component focuses on assessing the necessity and perceived effectiveness of the developed system in terms of functionality and usability. Quantitative data is collected through a structured pre-survey questionnaire utilizing a Dichotomous Scale (Yes/No). This instrument is designed to establish a clear consensus among residents and officials regarding whether the features of BrgyAlert effectively address existing reporting gaps. The data gathered is analyzed using descriptive statistical tools, specifically frequency distribution and percentage.

The research is conducted within the established capstone timeframe, during which the mobile prototype undergoes controlled testing and evaluation. This design ensures that all processes—from requirement gathering to AI model training—adhere to the standards set by the Data Privacy Act of 2012 (RA 10173), maintaining the confidentiality and security of the residents' sensitive information.

Formula for Percentage

The percentage of responses for each item will be calculated using this formula: P=fn×100P=nf ×100 Where: PP = Percentage ff = Frequency (number of "Yes" or "No" responses) nn = Total number of respondents

System Development Methodology

The researchers will employ the Agile Development Methodology to govern the creation and refinement of the BrgyAlert system. Agile is an iterative approach that breaks down the development process into manageable cycles or "sprints," allowing for continuous integration of feedback and technical adjustments (Beck et al., 2001). This methodology is particularly advantageous for BrgyAlert because it accommodates the complexities of training an AI risk stratification model while simultaneously building a user-centric mobile interface. As observed by Schwaber and Sutherland (2020), this framework is iterative and thus allows teams to manage unpredictable requirements and provide high-quality software through continuous inspection and adaptation. This framework ensures that the final output of the study is not only technically feasible but also in the context of the operational realities of incident management in Barangay Lepa.

Phase 1: Requirements Gathering and Analysis

The researchers will perform a thorough analysis of the state of incident reporting in Barangay Lepa during this first phase. A dual-track analysis is required: first, identifying the administrative pain points and challenges within the barangay's current manual incident recording and reporting practices as experienced by the residents; and second, ensuring the technical parameters of the system align with the legal frameworks of RA 9262 and RA 10173 (Data Privacy Act of 2012). In order to produce a thorough Software Requirements Specification (SRS), the researchers will record the "User Stories" (such as how a resident reports an incident via mobile versus how an official views the analytics dashboard). As the project's cornerstone, this phase ensures that the AI's classification logic and the system's features are based on actual community requirements and identified reporting gaps.

Phase 2: System Design and Architecture

The analyzed requirements will be converted into high-level technical blueprints. This includes the development of a System Architecture that illustrates the data flow between the mobile application used by residents, the cloud-based server, and the AI engine. The design process also involves creating the Entity Relationship Diagram (ERD) to ensure the database can securely store incident types, timestamps, and responder logs. On the front-end, the researchers will develop UI/UX wireframes with a "minimalist and high contrast" design, ensuring that any resident—even those in high-stress situations or low-light environments—can navigate the app and report an incident with minimal friction.

Phase 3: Technical Development

At this stage, the design is transformed into a working product through two parallel tracks: Mobile Development and AI Architecture. The mobile track involves coding the front-end interface and integrating the backend API for real-time data synchronization. Concurrently, the AI track focuses on developing the Risk Stratification algorithm. The researchers will train the model using a dataset of incident keywords (e.g., "emergency," "theft," "medical assistance") to identify situational urgency. This phase is highly iterative, as the model is continuously adjusted until it achieves a high degree of accuracy in distinguishing high-priority emergencies from standard administrative inquiries.

Phase 4: Testing and Quality Assurance

The prototype will undergo a stringent testing phase to ensure reliability. The researchers will perform Unit Testing for individual features (such as the SOS button) and System

Integration Testing to verify that the AI correctly processes and categorizes reports sent from the mobile app. Validation of the AI's risk assessment is critical, involving simulated scenarios to ensure no life-threatening incident is misclassified. Additionally, the researchers will conduct a User Acceptance Test (UAT) with selected residents of Barangay Lepa to evaluate the system based on the Technology Acceptance Model (TAM), specifically measuring its perceived ease of use and perceived usefulness.

Phase 5: Deployment and Implementation

The final stage marks the transition from development to a live operational setting in Barangay Lepa. This includes distributing the mobile application to residents and installing the centralized Analytics Dashboard at the Barangay Hall. The researchers will hold orientation sessions for barangay officials and first responders to ensure a seamless transition from the manual system. A "Parallel Deployment" approach will be employed, where the digital system operates alongside the manual blotter for a brief period to ensure data consistency. Once stability is confirmed in the live environment, the BrgyAlert platform will be officially fully deployed for community use.

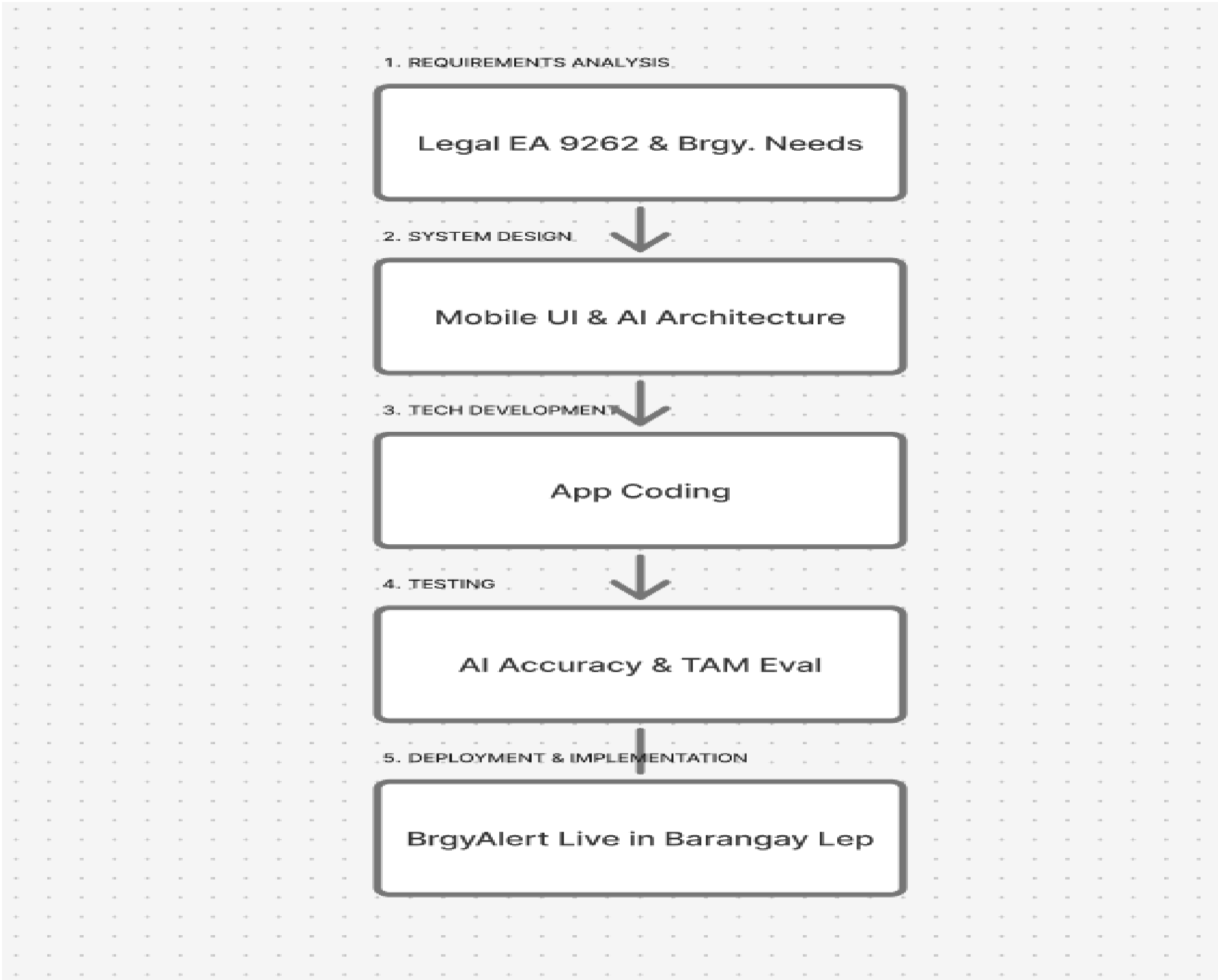

Figure 3.1: Brgyalert. System Development Life Cycle (Agile Model)

Figure 3.1 shows the iterative stages of the Agile Development Methodology as modified for the BrgyAlert system. To guarantee the accuracy of the AI risk stratification model, the process starts with requirements analysis, moves on to system design, and then enters an iterative cycle of technical development and testing. The system's deployment and implementation in Barangay Lepa marks the end of the cycle. It begins with a strong focus on gathering requirements based on local barangay needs and legal norms, then moves on to the mobile application design phase.

The "Agile Loop," which is represented by the middle portion, involves repeated technical development and testing to guarantee the accuracy of the AI's risk assessment prior to the system's eventual deployment and integration into the barangay's regular operations.

Requirement Analysis

The requirements analysis phase serves as the diagnostic foundation for the development process, translating Barangay Lepa's specific operational needs into technological specifications. Bridging the gap between the suggested AI-driven solution and the constraints of manual recording is the main goal of this stage. The researchers determined the crucial functional, non-functional, and technological requirements of the BrgyAlert system in order to guarantee that the application satisfies the particular mobility and safety needs of the community.

3.1.1 Functional and Non-Functional Requirements

The functional requirements specify the exact activities and behaviors that the BrgyAlert system must do in order to meet user expectations. The Incident Reporting Module, which allows victims or witnesses to enter vital incident facts including exact geographical information and the particular type of violence, is at the heart of the platform. The system incorporates an AI Risk Stratification engine that uses machine learning algorithms to automatically classify incoming reports into High, Medium, or Low risk levels in order to maximize the barangay's response. Additionally, an automated notification system guarantees that high-risk situations are promptly identified and forwarded to the relevant authorities, and an administrative dashboard gives officials a centralized interface for real-time analytics and case management.

The non-functional criteria outline the qualities and limitations that guarantee the efficiency and dependability of the system. First and foremost is data security, which places a high priority on encrypting critical data to stop record tampering and illegal access. Another important consideration is usability; the system's mobile-optimized UI makes it easy for users to use the platform even under stressful circumstances. Furthermore, the system is designed with great reliability in mind, guaranteeing round-the-clock accessibility to handle emergency reports at any time of day.

3.1.2 Software Requirements

The software requirements specify the programming tools and digital environment required for the system's essential features. To guarantee compatibility with different smartphone operating systems, the mobile application needs a cross-platform development framework (like Flutter or React Native). A secure cloud server is required by the backend architecture to manage databases and implement the AI risk stratification model. Finally, specialized machine learning libraries are used in the construction of the AI Engine. These libraries are crucial for training and improving the risk classification models that power the system's predictive skills.

3.1.3 Hardware Requirements

The hardware requirements outline the actual gadgets required to bridge the gap between the community and barangay responders. High-performance cellphones are the main piece of equipment used by barangay officials to get alerts and by constituents to report. A specialized computer workstation with a multi-core processor and enough RAM is needed to host the administrative dashboard for the centralized monitoring station in Barangay

Lepa. Furthermore, the real-time transfer of data from the field to the AI engine requires a dependable network infrastructure, such as Wi-Fi routers or mobile data modules.

3.1.4 Data Requirements

The data requirements specify the precise information required to power the system's evaluation and decision-making procedures. The AI's risk classification is trained using primary incident data, which includes victim demographics, the type of offense, and situational descriptions. Geographical information is also needed by the system to precisely identify the sites of incidents. Last but not least, the system's performance in terms of security, usability, and functional suitability must be measured using evaluation data gathered from Barangay Lepa authorities and citizens.

Required Documentation

The researchers are dedicated to keeping an extensive collection of records in order to guarantee the BrgyAlert system's sustainability and traceability. The System Requirements Specification (SRS), which outlines all of the functional and technological blueprints created during this stage, is one example of this. Agile documentation, such as sprint backlogs and progress reports, will be kept up to date throughout the development cycles to monitor the advancement of mobile features and AI models. The Barangay Lepa authorities will get a User Manual and Technical Documentation after the study is finished, which will act as a guide for system operation, troubleshooting, and upcoming software updates.

Software Requirements

Category	Specification
OS	- Windows 10/ Windows 11 (Development) - Android 8.0+ / iOS 14+ (Target Mobile Runtime)
Application	- Visual Studio Code - Node.js & npm (v18+ LTS) - Expo CLI & Metro Bundler - Firebase CLI - Expo Go (Mobile testing app)

Database	- Cloud Firestore (Serverless NoSQL) - Firebase Authentication - Firebase Cloud Storage - Semaphore.co SMS API

Table 1. Software Requirements

Table 1 presents the essential requirements identified during the system's development process. The Software Requirements Specification (SRS) serves as a detailed reference that defines the project's objectives.

Hardware Requirements

Category	Specification
Dekstop/Laptop	- Processor: Ryzen 5 5600G (or equivalent) - RAM: 16 GB RAM - Hard Disk: 500 GB hard disk / SSD

Output Devices	- Monitor - Mouse - Keyboard - Alarm - Tablet
Mobile Devices	- Android or IOS Smartphones (citizen & responders) - Integrated GPS Module - Camera Module - Active SIM Card
Internet/Ethernet Connection	At least 150 Mbps

Table 2. Hardware Requirements

Table 2 presents the hardware components utilized in the development of the BrgyAlert Incident Reporting & Analytics Platform. It outlines the specifications of the devices used during the system's programming and testing phases.

Data Requirements

Category	Specification
Preliminary Survey Questionnaire	Conducted at the initial stage of the gathering essential local data, user challenges, and feedback from respondents in Barangay Lepa.
Evaluation	Structured assessment of the developed system where empirical data is analyzed to provide insights for system quality.
User Account Records	Profile datasets containing User UIDs, full names, emails, contact numbers, and RBAC roles ("citizen" or "admin") to manage authentication logs.
Incident Alerts & Logs	Structured datasets containing alertIds, reporter names, category details (Fire, Flood, Medical, Vioolence etc.), status,

	urgency level, image paths, and timestamps.
Geographical Coordinates	High-precision GPS latitude and longitude mappings to visualize incidents on Map APIs.
Compressed SMS Payloads	String-delimited datasets formatted as "BA![Cat]![Landmark]![Details]" for offline storage and parsing by the SMS gateway.

Table 3. Data Requirements

Table 3 presents the data requirements essential for the development of the BrgyAlert Incident Reporting & Analytics Platform. These data inputs are necessary to ensure the system's effectiveness in managing and organizing records efficiently.

Instrumentation

The researchers developed a primary data collection instrument to gather empirical data for the development and evaluation of BrgyAlert. This instrument is a structured pre-survey questionnaire designed to assess the current incident reporting landscape in Barangay Lepa and determine the potential impact of the proposed digital solution. The tool is strategically divided into three major parts to ensure a comprehensive analysis of the community's needs and the system's viability.

The first part of the instrument focuses on the Respondent Profile, which collects demographic data including the name, age, sex, civil status, occupation, and length of residency of the participants. This section is vital for ensuring that the feedback is coming from a representative sample of the community stakeholders and helps in contextualizing the user experience across different demographic groups.

The second part, titled Current Incident Reporting Experience, utilizes a dichotomous scale to evaluate the efficiency of existing reporting practices in the barangay. It specifically identifies common pain points such as delays in emergency response, the lack of status updates for reported incidents, and overall satisfaction with manual methods. The data gathered from this section serves as the baseline evidence for the study, highlighting the technical and operational gaps that BrgyAlert aims to bridge.

The third part, Features and Effectiveness of BrgyAlert, assesses the participants' perceptions of the proposed system’s potential benefits. This section explores key functional areas such as the improvement of communication between residents and officials, the efficiency of digital record management, and the overall enhancement of community safety. To maintain research integrity and ethical standards, a confidentiality clause is included at the end of the instrument, ensuring that all responses are handled in strict accordance with the Data Privacy Act of 2012 and used solely for academic purposes.

Toold For Data Analysis

To evaluate the effectiveness of the system and ensure the accuracy of the results, the researchers utilized descriptive statistical tools to process the data gathered from the 110 respondents. The data analysis serves as a guide to determine how the developed platform addresses the operational gaps in Barangay Lepa.

The following statistical measures were employed:

Frequency and Percentage Distribution. This tool was used to summarize the numerical data derived from the survey questionnaires. It specifically measures the distribution of responses regarding the current challenges in manual reporting and the perceived usefulness of the digital features of BrgyAlert.

The formula used for the calculation is:

P = (f/n) * 100

Where: **P**=Percentage **f**=Frequency of a specific response **n**=Total number of respondents (110)

Ranking. This measure was used to prioritize the common problems identified in the existing manual blotter system and to categorize the most anticipated features of the platform. By ranking the frequencies, the researchers were able to align the system's risk stratification and SMS alert capabilities with the actual needs of the community.

Technical Evaluation and Software Standard. While the initial data was gathered through user feedback, the overall quality and performance of the system were also assessed to ensure it met essential criteria for reliability and speed.

All gathered quantitative data were tabulated and analyzed using Microsoft Excel, providing a clear and measurable evaluation of the system's performance as perceived by its end users.

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